

Homework — 1.4.3 Boolean Algebra

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.4.3 lesson. *SHOW ALL WORKING.*

Part A — This week's topic (~14 marks)

1. Complete the truth table for the expression $Q = (A + B) \cdot \neg C$. [4] (AO2)

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A	B	C	A+B	¬C	Q
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

2. Use **De Morgan's law** (and double negation where needed) to simplify $\neg(A \cdot \neg B)$ to an expression with no outer bar. Show each step. [3] (AO2)

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Working space:

3. Simplify the expression $A + \neg A \cdot B$ as far as possible, and name the law(s) used. [3] (AO1/AO2)

Working space:

4. State the Boolean expressions for the **Sum** and **Carry** outputs of a **half adder**, and explain why a half adder **cannot** be used on its own to add the middle columns of a multi-bit binary addition. [2] (AO1)

Working space:

5. State **two** differences between **combinational** logic and **sequential** logic, giving one example component of each. [2] (AO1)

Working space:

Part B — Retrieval: keep it fresh (~6 marks)

6. Explain the difference between **symmetric** and **asymmetric** encryption, and state **one** advantage of asymmetric encryption. (1.3) [3] (AO1)

Working space:

7. Describe how the **round robin** CPU scheduling algorithm works, and state **one** advantage of it. (1.2) [3] (AO1)

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Working space:
